

4 Integer Linear Programming Problems (ILPP): Branch and Bound Method

4.1 Introduction

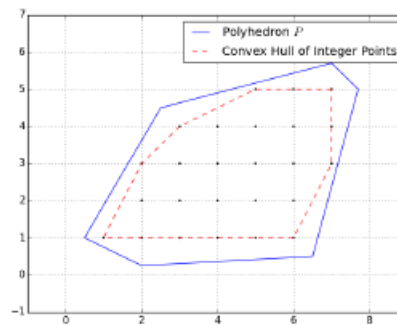
- An **integer linear optimization problem** (ILP) is the same as a linear optimization problem except that the variables can take on only **integer** values.
- If only some of the variables are constrained to take on integer values, then we call the program a **mixed integer linear optimization problem** (MILP).
- Why do we need integer variables?
- If the variable is associated with a physical entity that is **indivisible**, then it must be integer. For example:
 - Shares of a stock.
 - Investments that can only be made in fixed amounts.

4.2 ILPP Model

- Generally, an ILPP model is presented as follows:

$$\begin{aligned} \text{Max/Min:} \quad & Z = c^T x \\ \text{subject to:} \quad & Ax = b, \\ & x \geq 0, \quad \text{and } x \text{ are integers} \end{aligned}$$

- The **feasible region** is the integer points inside a polyhedron.



- For example, this is a simple linear programming problem (LPP).

$$\begin{aligned} \text{Maximize:} \quad & Z = 7x_1 + 9x_2 \\ \text{subject to:} \quad & -x_1 + 3x_2 \leq 6 \\ & 7x_1 + x_2 \leq 35 \\ & x_1, x_2 \geq 0, \quad x_1, x_2 \end{aligned}$$

- If, however, we add the condition that x_1 and x_2 are also integer then this problem is termed as an integer linear programming (ILP) problem.

$$\begin{aligned} \text{Maximize: } & Z = 7x_1 + 9x_2 \\ \text{subject to: } & -x_1 + 3x_2 \leq 6 \\ & 7x_1 + x_2 \leq 35 \\ & x_1, x_2 \geq 0, \quad x_1, x_2 \text{ are integer} \end{aligned}$$

- Methods of solving LPPs such as **Graphical method** and **Simplex method** can no longer help us find optimum solution to the above ILP.
- Two general methods have been developed for solving ILPs: (1) Branch-and-Bound (B & B) method (2) Cutting-Plane method. In this unit, we shall focus on the B & B method.

4.3 Branch and Bound (B & B) Method

- This technique is applicable to both all integer programming problems (ILPP) as well as mixed integer programming problems (MILPP).
- In it, we decompose the given problem into sub-problems.
- The B & B approach is based on the principle that the total set of feasible solutions can be partitioned into smaller subsets of solutions.
- These smaller subsets can then be evaluated systematically until the best solution is found.
- When B & B approach is applied to an integer programming problem, it is used in conjunction with the normal noninteger solution approach.
- The steps (**Algorithm**) of the branch and bound method for determining an optimal integer solution for a **maximization model** (with $\leq$ constraints) can be summarized as follows:
 1. Find the optimal solution to the linear programming model with the integer restrictions relaxed.
 2. At node 1 let the relaxed solution be the upper bound and the rounded-down integer solution be the lower bound.
 3. **Select the variable with the greatest fractional part for branching.** Create two new constraints for this variable reflecting the partitioned integer values. The result will be a new $\leq$ constraint and a new $\geq$ constraint.
 4. Create two new nodes, one for the $\leq$ constraint and one for the $\geq$ constraint.
 5. Solve the relaxed linear programming model with the new constraint added at each of these nodes.
 6. The relaxed solution is the upper bound at each node, and the existing maximum integer solution (at any node) is the lower bound.
 7. If the process produces a feasible integer solution with the greatest upper bound value of any ending node, the optimal integer solution has been reached. If a feasible integer solution does not emerge, branch from the node with the greatest upper bound.

8. Return to step 3.

For a **minimization model**, relaxed solutions are rounded up, and upper and lower bounds are reversed.

In summary,

1. Relax the solution space of the ILP by deleting the integer restriction on all integer variables. The result of the relaxation is a regular LP.
2. Solve the LP, and identify its continuous optimum.
3. Starting from the continuous optimum point, add special constraints that iteratively modify the LP solution space in a manner that eventually renders an optimum extreme point satisfying the integer requirements.

Example 1:

Use Branch and Bound technique to solve the following integer programming problem.

$$\begin{array}{ll}\text{Maximize:} & Z = 7x_1 + 9x_2 \\ \text{subject to:} & -x_1 + 3x_2 \leq 6 \\ & 7x_1 + x_2 \leq 35 \\ & x_1, x_2 \geq 0, \quad x_1, x_2 \text{ are integer}\end{array}$$

Solution:

Step 1: Relax the integer restrictions on variables x_1 and x_2 . Use graphical method to solve the resulting LP1.

$$\begin{array}{ll}\text{Maximize:} & Z = 7x_1 + 9x_2 \\ \text{subject to:} & -x_1 + 3x_2 \leq 6 \dots\dots(1) \\ & 7x_1 + x_2 \leq 35 \dots\dots(2) \\ & x_1, x_2 \geq 0\end{array}$$

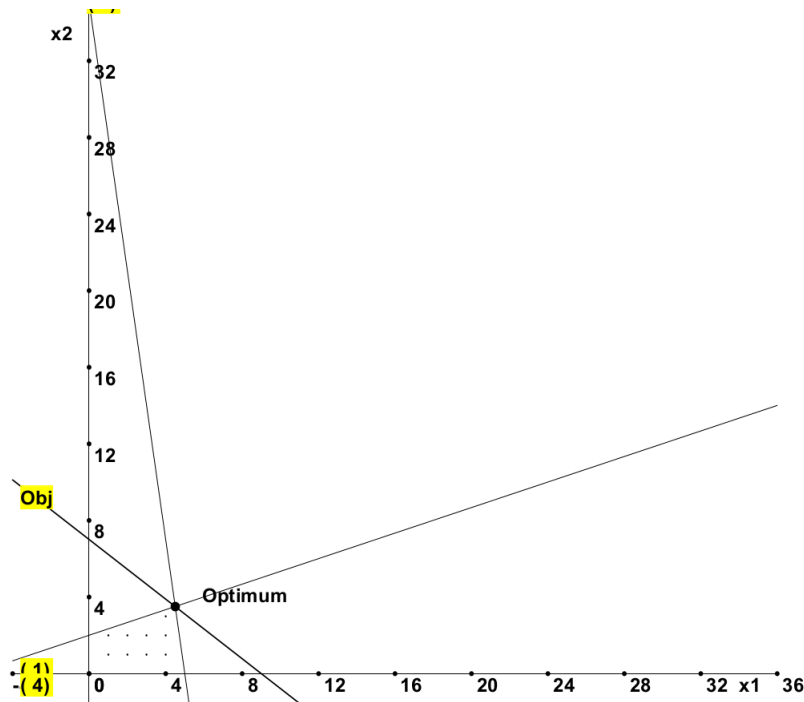


Fig. 4

The optimum solution of LP1 is $x_1 = 4.5, x_2 = 3.5$ and $z = 63$.

Note: $z = 63$ is called the **upper bound** - we do not expect to get any better value of z than this value.

Step 2: Because the optimum LP1 solution does not satisfy the integer restrictions, the solution space is subdivided in a systematic manner that eventually locates the ILP optimum. B & B selects a **non-integer** value. If both are non-integer values then **select the one with largest decimal portion**. Here, both variables have the same fractional part, i.e., 0.5. Choose any one (choose arbitrarily). Let us choose x_1 as the **branching variable**. Selecting $x_1 (= 4.5)$, the region $4 < x_1 < 5$ contain no integer values and can be deleted. The original LP1 is therefore decomposed into two new LPs (sub-problems):

$$\text{LP2 space} = \text{LP1} + (x_1 \geq 5)$$

$$\text{LP3 space} = \text{LP1} + (x_1 \leq 4)$$

That is LP2 is:

$$\begin{aligned} \text{Maximize:} \quad & Z = 7x_1 + 9x_2 \\ \text{subject to:} \quad & -x_1 + 3x_2 \leq 6 \dots\dots(1) \\ & 7x_1 + x_2 \leq 35 \dots\dots(2) \\ & x_1 \geq 5 \dots\dots(3) \\ & x_1, x_2 \geq 0. \end{aligned}$$

We proceed to solve LP3.

$$\begin{aligned} \text{Maximize:} \quad & Z = 7x_1 + 9x_2 \\ \text{subject to:} \quad & -x_1 + 3x_2 \leq 6 \text{(1)} \\ & 7x_1 + x_2 \leq 35 \text{(2)} \\ & x_1 \leq 4 \text{(3)} \\ & x_1, x_2 \geq 0. \end{aligned}$$

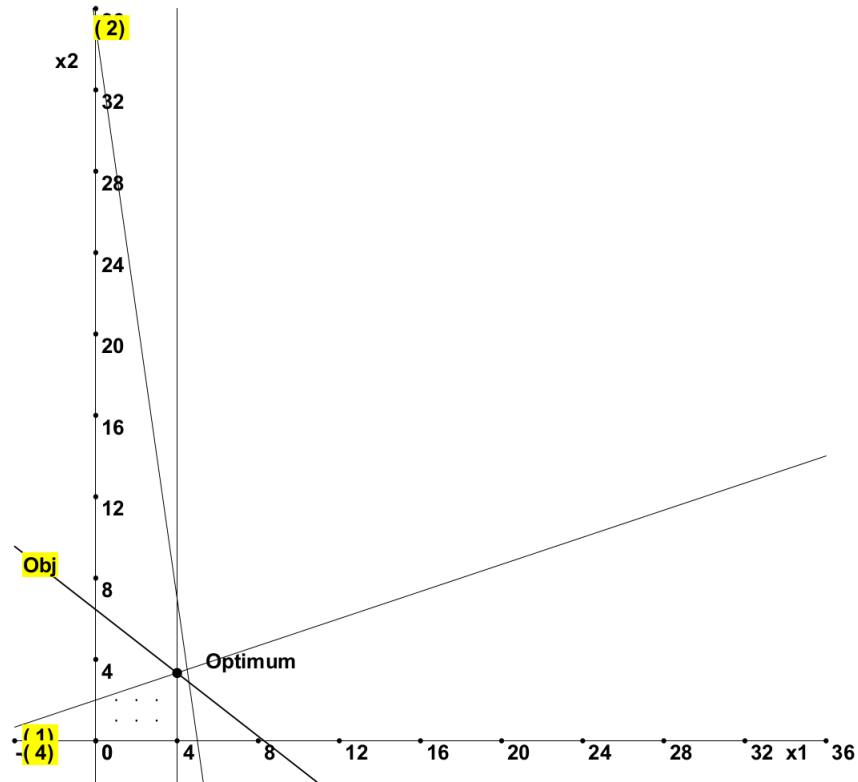


Fig. 6

The solution of LP3 is $x_1 = 4, x_2 = 10/3$ and $z = 58$.

Step 5: Since $x_2 (= 10/3)$ is not an integer, we subdivide x_2 into $(x_2 \leq 3 \text{ and } x_2 \geq 4)$. This results into LP4 and LP5 respectively.

$$\text{LP4 space} = \text{LP3} + (x_2 \geq 4)$$

$$\text{LP5 space} = \text{LP3} + (x_2 \leq 3)$$

Step 6: Using graphical method, we observe that LP4 has **No feasible solution**.

Step 7: Similarly, solve LP5. LP5 is given as

$$\begin{aligned} \text{Maximize:} \quad & Z = 7x_1 + 9x_2 \\ \text{subject to:} \quad & -x_1 + 3x_2 \leq 6 \dots\dots\dots(1) \\ & 7x_1 + x_2 \leq 35 \dots\dots\dots(2) \\ & x_1 \leq 4 \dots\dots\dots(3) \\ & x_2 \leq 3 \dots\dots\dots(4) \\ & x_1, x_2 \geq 0. \end{aligned}$$

Using graphical method, we observe that LP5 has solutions: $x_1 = 4, x_2 = 3$ and $z = 55$. Notice that both x_1 and x_2 are integers in LP5 and $z=55$ is maximum. We thus conclude that the optimum ILP solution is: $x_1 = 4, x_2 = 3$ and $z = 55$. [The Best Lower Bound]

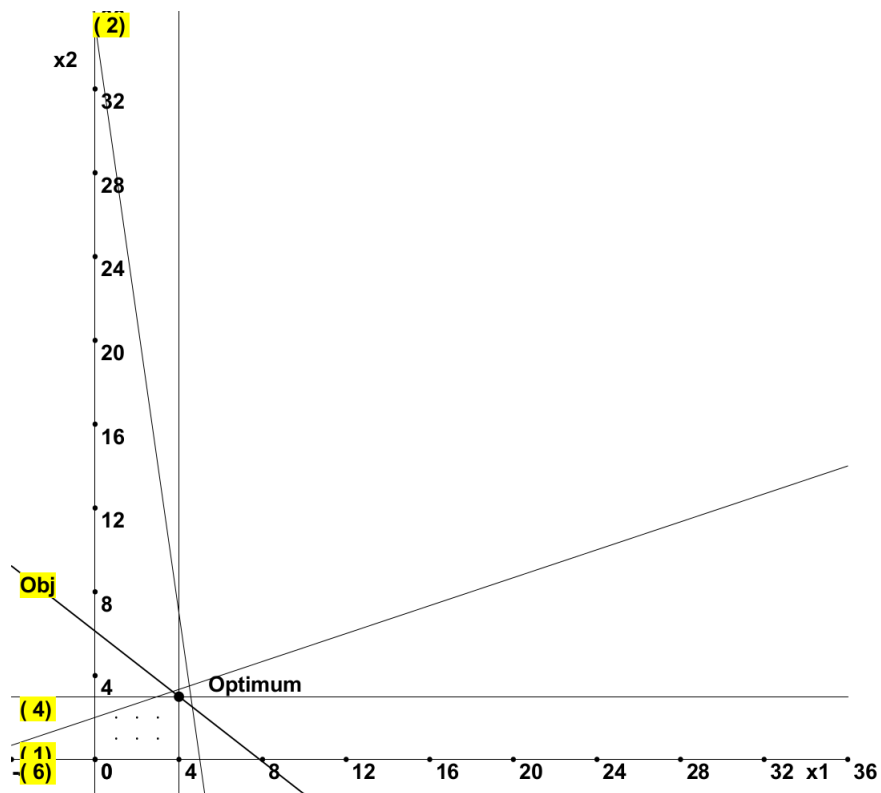


Fig. 7

The associated B & B **tree diagram** for the above ILP is given by

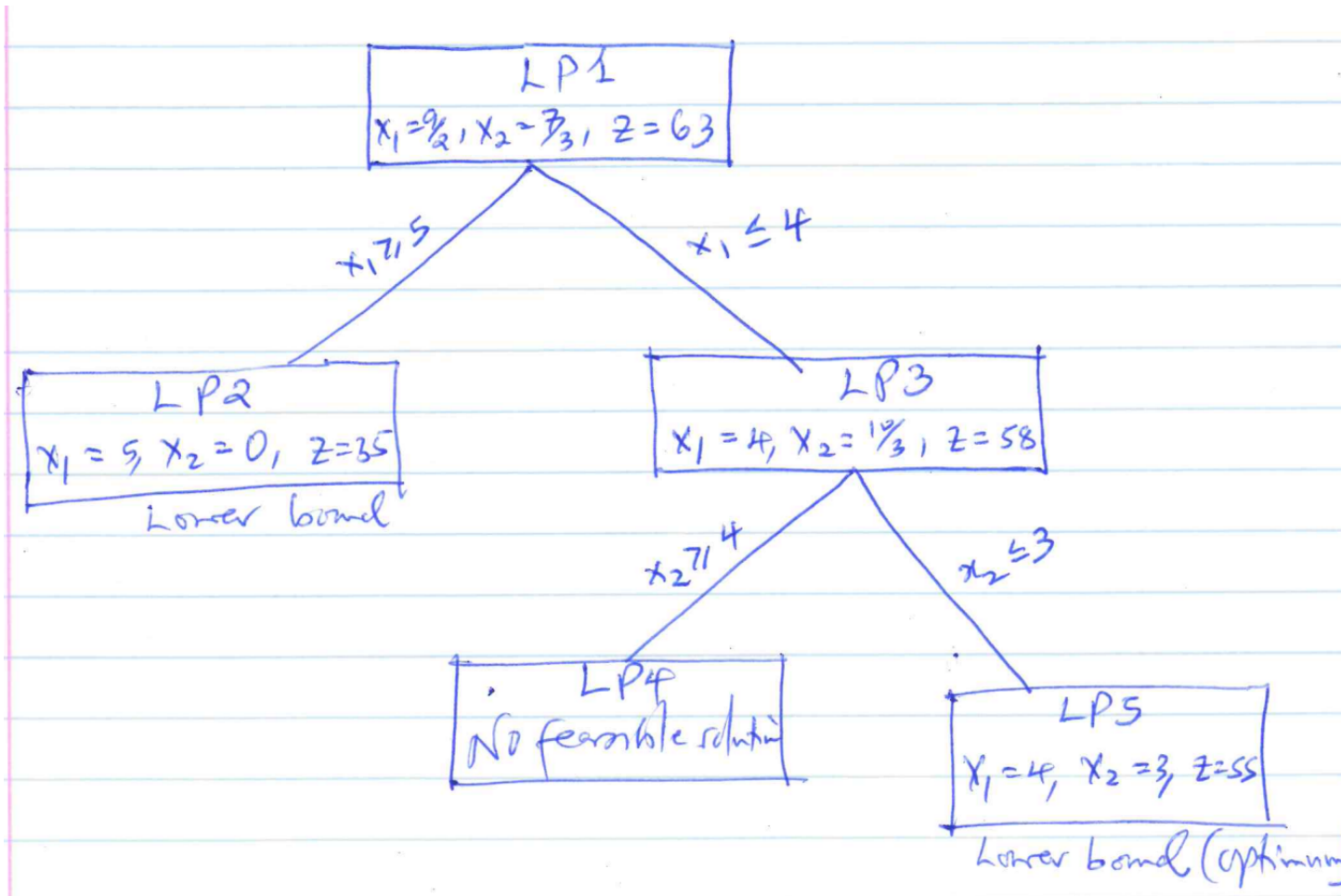


Fig. 8: B & B Tree for example 1

Verify your results using TORA software.

TORA Moment: Integer Programming B& B Algorithm

TORA can be used to generate iterations of the B & B Algorithm. From **Main Menu**, select **Integer programming** $\Rightarrow$ **Input the ILPP data**. Next, from **SOLVE/MODIFY** menu, select **solve problem** $\Rightarrow$ **Go to output screen**. In the output screen, select a **Automated B & B**, [for optimal solution only] OR select **User-guided B & B** [for **All iterations** leading to B & B tree diagram. Select the branching nodes as directed on the top left corner of the output screen.

Example 2:

Develop the B & B tree for the following ILPP. Write down the optimal solution.

$$\begin{aligned}
 \text{Maximize:} \quad & Z = 5x_1 + 6x_2 \\
 \text{subject to:} \quad & x_1 + x_2 \leq 5 \\
 & 4x_1 + 7x_2 \leq 28 \\
 & x_1, x_2 \geq 0, \quad x_1, x_2 \text{ are integer}
 \end{aligned}$$

Solution:

Step 1: Relax the integer restrictions on variables x_1 and x_2 . Use graphical method to solve the resulting LP1.

$$\begin{aligned}
 \text{Maximize:} \quad & Z = 5x_1 + 6x_2 \\
 \text{subject to:} \quad & x_1 + x_2 \leq 5 \dots\dots\dots(1) \\
 & 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

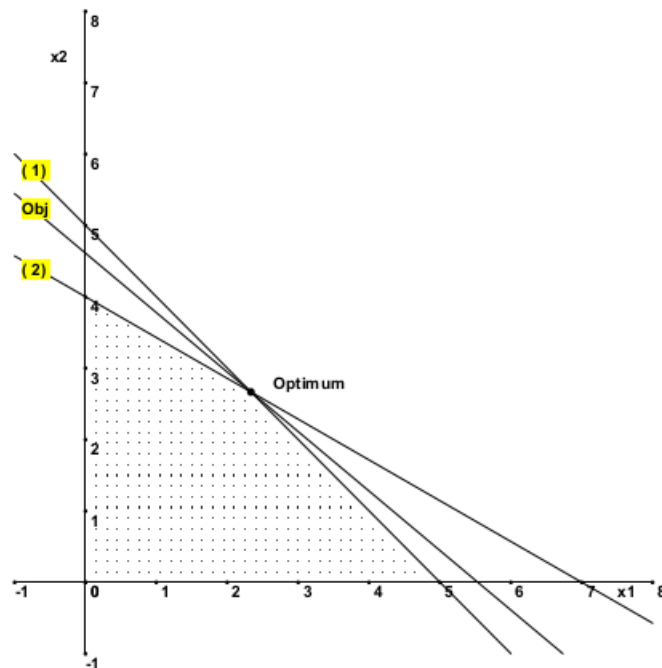


Fig. 9

The optimum solution of LP1 is $x_1 = 2.33$, $x_2 = 2.67$ and $z = 27.67$

Step 2: Here, x_2 (with larger decimal portion) is the **branching variable**. The original LP1 is therefore replaced with two new LPs:

$$\text{LP2 space} = \text{LP1} + (x_2 \leq 2)$$

$$\text{LP3 space} = \text{LP1} + (x_2 \geq 3)$$

That is LP2 is:

$$\begin{aligned} \text{Maximize: } & Z = 5x_1 + 6x_2 \\ \text{subject to: } & x_1 + x_2 \leq 5 \dots\dots\dots(1) \\ & 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\ & x_2 \leq 2 \dots\dots\dots(3) \\ & x_1, x_2 \geq 0 \end{aligned}$$

Step 3: Solve LP2 by graphical method. The solution of LP2 is $x_1 = 3, x_2 = 2$ and $z = 27$. The LP2 solution satisfies the integer requirements for x_1 and x_2 . Hence, LP2 is said to be **fathomed**. $z = 27$ is considered a **lower bound** (LB^*).

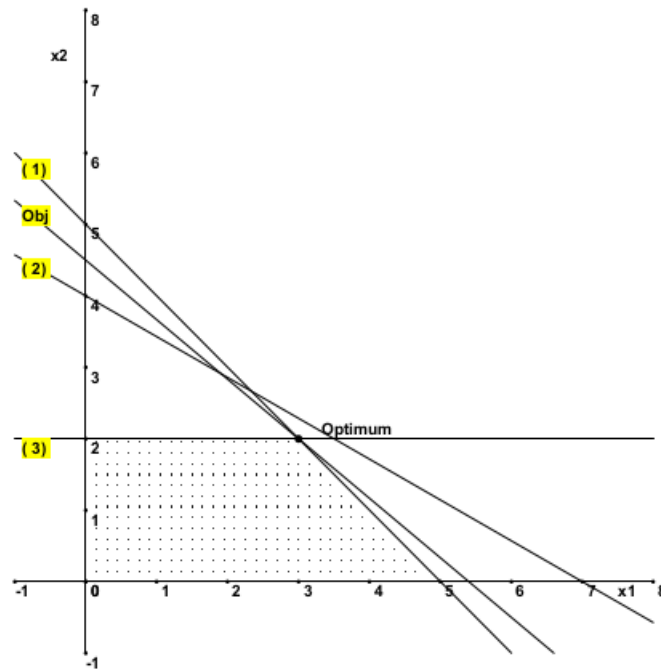


Fig. 10

Step 4: Solve LP3 by graphical method too. LP3 is given as:

$$\begin{aligned} \text{Maximize: } & Z = 5x_1 + 6x_2 \\ \text{subject to: } & x_1 + x_2 \leq 5 \dots\dots\dots(1) \\ & 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\ & x_2 \geq 3 \dots\dots\dots(3) \\ & x_1, x_2 \geq 0 \end{aligned}$$

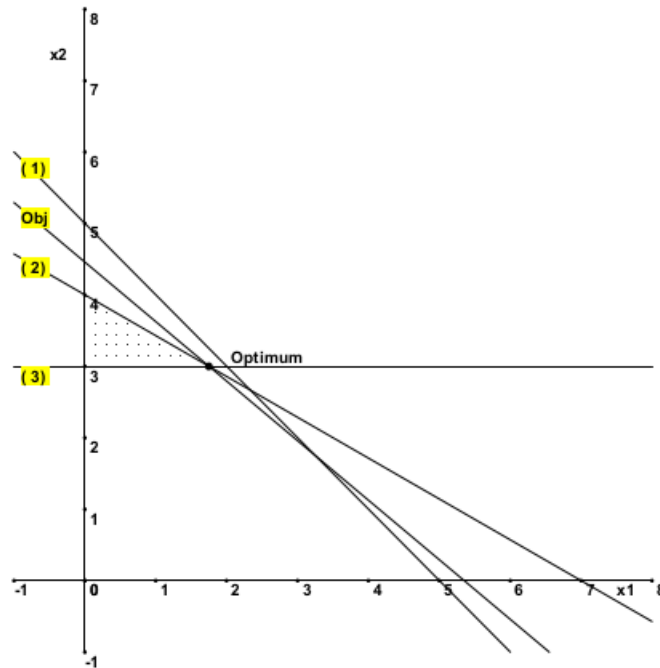


Fig. 11

The solution of LP3 is $x_1 = 1.75$, $x_2 = 3$ and $z = 26.75$.

Step 5: Since $x_1 (= 1.75)$ is not an integer, we subdivide x_1 into $(x_1 \leq 1$ and $x_1 \geq 2)$. This results into LP4 and LP5 respectively.

$$\text{LP4 space} = \text{LP3} + (x_1 \geq 2)$$

$$\text{LP5 space} = \text{LP3} + (x_1 \leq 1)$$

Step 6: LP4 is given as follows:

$$\begin{aligned}
&\text{Maximize:} && Z = 5x_1 + 6x_2 \\
&\text{subject to:} && x_1 + x_2 \leq 5 \dots\dots\dots(1) \\
& && 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\
& && x_2 \geq 3 \dots\dots\dots(3) \\
& && x_1 \geq 2 \dots\dots\dots(4) \\
& && x_1, x_2 \geq 0
\end{aligned}$$

Using graphical method, we observe that LP4 has **No feasible solution**.

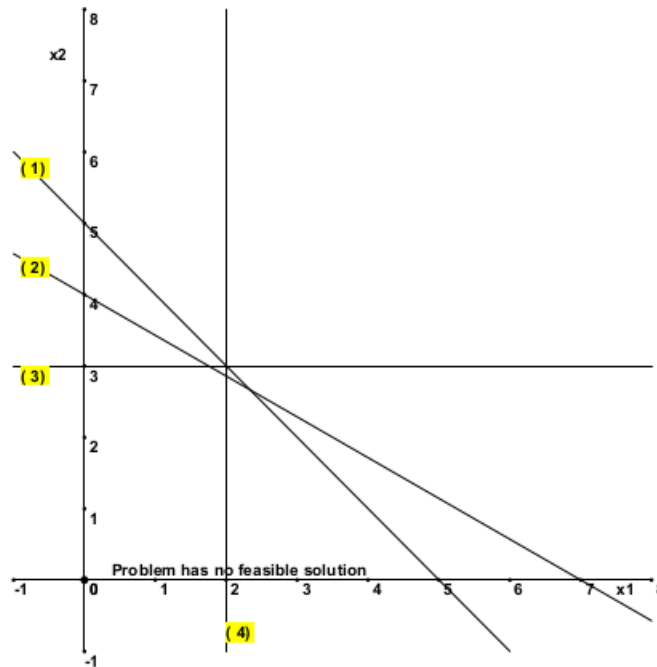


Fig. 12

Step 7: Similarly, solve LP5. LP5 is given as:

$$\begin{aligned}
&\text{Maximize:} && Z = 5x_1 + 6x_2 \\
&\text{subject to:} && x_1 + x_2 \leq 5 \dots\dots\dots(1) \\
& && 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\
& && x_2 \geq 3 \dots\dots\dots(3) \\
& && x_1 \leq 1 \dots\dots\dots(4) \\
& && x_1, x_2 \geq 0
\end{aligned}$$

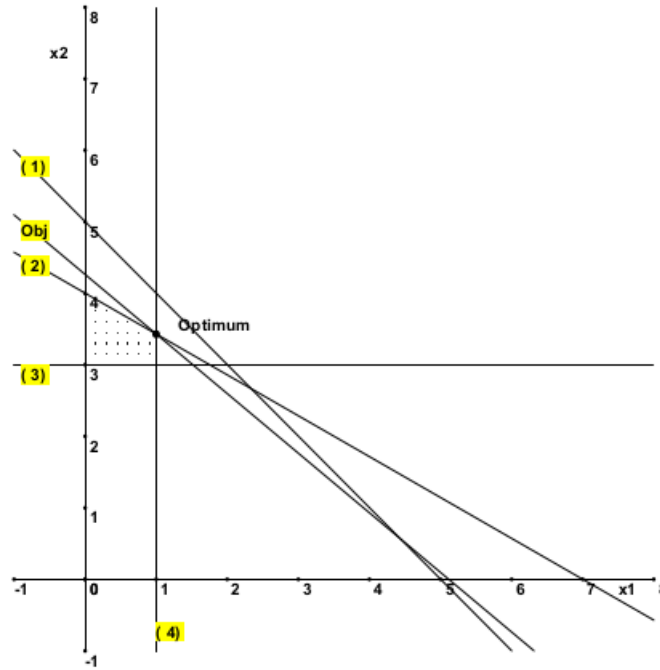


Fig. 13

LP5 has solutions: $x_1 = 1, x_2 = 3.42$ and $z = 25.57$.

Step 8: Notice that x_2 is not an integer and we subdivide x_2 into $(x_2 \leq 3$ and $x_2 \geq 4)$. This results into LP6 and LP7 respectively.

$$\text{LP6 space} = \text{LP5} + (x_2 \leq 3)$$

$$\text{LP7 space} = \text{LP5} + (x_2 \geq 4)$$

LP6 is given as:

$$\begin{aligned} \text{Maximize:} \quad & Z = 5x_1 + 6x_2 \\ \text{subject to:} \quad & x_1 + x_2 \leq 5 \dots\dots\dots(1) \\ & 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\ & x_2 \geq 3 \dots\dots\dots(3) \\ & x_1 \leq 1 \dots\dots\dots(4) \\ & x_2 \leq 3 \dots\dots\dots(5) \\ & x_1, x_2 \geq 0 \end{aligned}$$

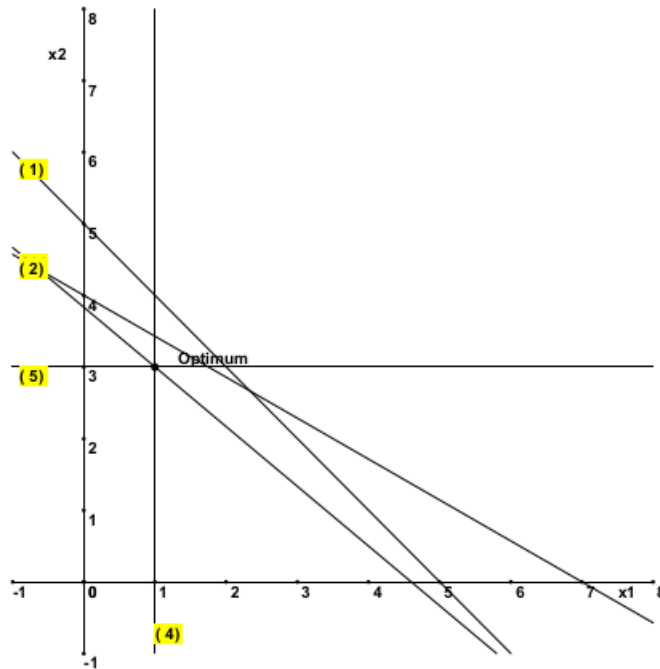


Fig. 14

The solution of LP6 is $x_1 = 1$, $x_2 = 3$ and $z = 23$. Hence, LP6 is said to be **fathomed**.

Step 9: Finally, solve LP7. LP7 is given as:

$$\begin{aligned}
 &\text{Maximize:} && Z = 5x_1 + 6x_2 \\
 &\text{subject to:} && x_1 + x_2 \leq 5 \dots\dots\dots(1) \\
 &&& 4x_1 + 7x_2 \leq 28 \dots\dots\dots(2) \\
 &&& x_2 \geq 3 \dots\dots\dots(3) \\
 &&& x_1 \leq 1 \dots\dots\dots(4) \\
 &&& x_2 \geq 4 \dots\dots\dots(5) \\
 &&& x_1, x_2 \geq 0
 \end{aligned}$$

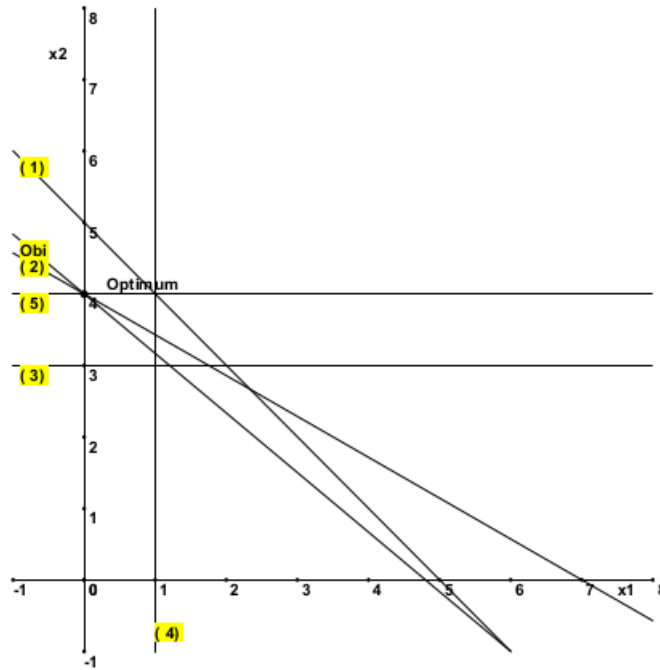


Fig. 15

The solution of LP7 is $x_1 = 0$, $x_2 = 4$ and $z = 24$. Hence, LP7 is also **fathomed**.

Upon comparing the integer solutions is LP2, LP6 and LP7, we observe that LP2 has the largest value of the objective function, z and hence forms the optimal solution to the given ILPP.

We thus conclude that the **optimum ILP solution is:** $x_1 = 3$, $x_2 = 2$ and $z = 27$.

The associated **B & B tree diagram** for the above ILP is given by

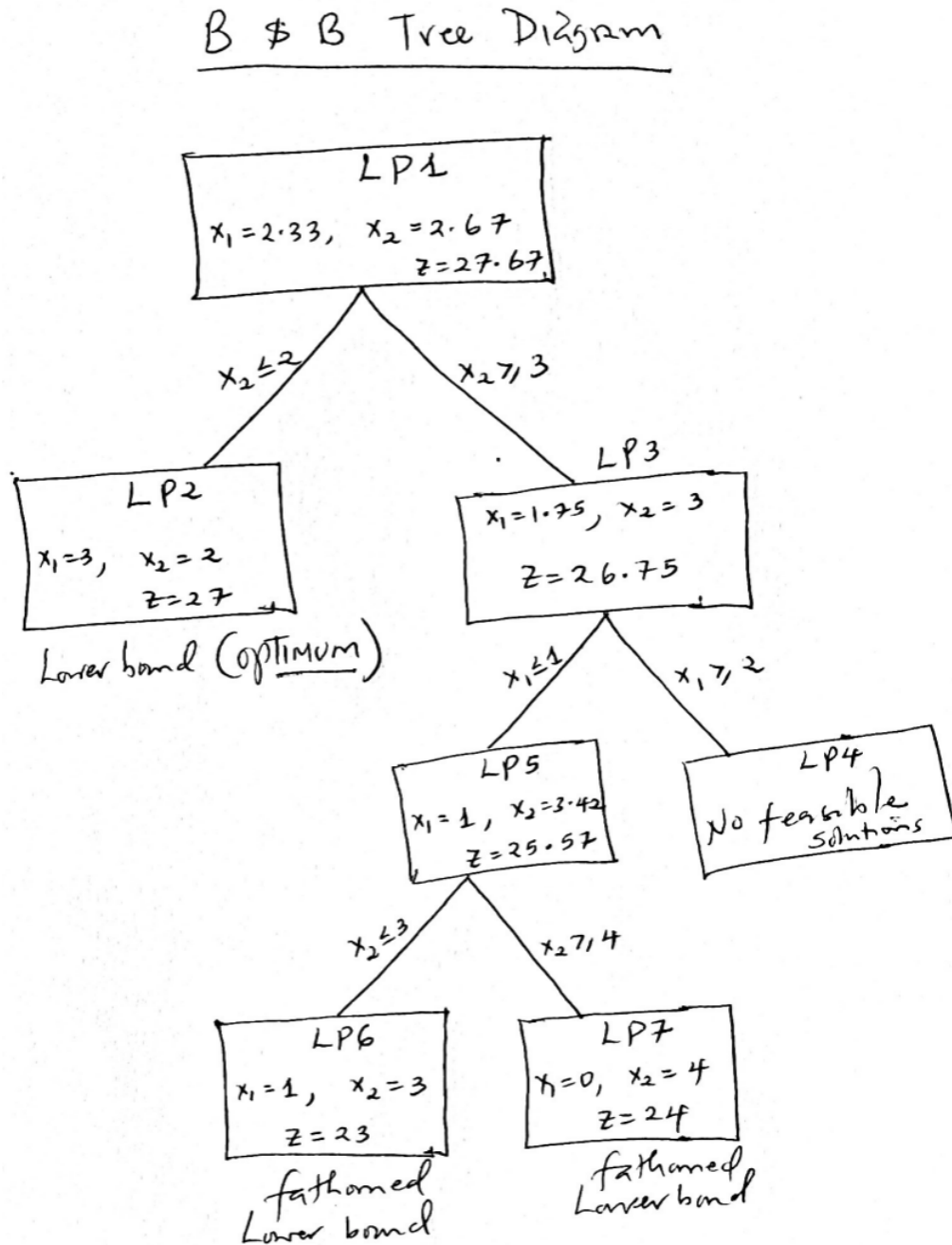


Fig. 16: B & B Tree for example 2

Exercise 7.

1. Develop the B & B tree for each of the following problems. Write down the optimal solution.

a. (based on Graphical method)

$$\begin{aligned} \text{Maximize:} \quad & Z = 5x_1 + 4x_2 \\ \text{subject to:} \quad & x_1 + x_2 \leq 5 \\ & 10x_1 + 6x_2 \leq 45 \\ & x_1, x_2 \geq 0 \quad \text{and integer} \end{aligned}$$

Optimal Solution: $x_1 = 3, x_2 = 2, Z = 23$

b. (based on TORA software)

$$\begin{aligned} \text{Maximize:} \quad & Z = 3x_1 + 2x_2 \\ \text{subject to:} \quad & 2x_1 + 5x_2 \leq 18 \\ & 4x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0 \quad \text{and integer} \end{aligned}$$

Optimal Solution: $x_1 = 4, x_2 = 1, Z = 14$

2. The owner of a machine shop is planning to expand by purchasing some new machines—presses and lathes. The owner has estimated that each press purchased will increase profit by \$100 per day and each lathe will increase profit by \$150 daily. The number of machines the owner can purchase is limited by the cost of the machines and the available floor space in the shop. The machine purchase prices and space requirements are as follows.

Machine	Required	
	Floor Space (ft^2)	Purchase Price
Press	15	\$ 8,000
Lathe	30	4,000

The owner has a budget of \$40,000 for purchasing machines and 200 square feet of available floor space. The owner wants to know how many of each type of machine to purchase to maximize the daily increase in profit.

Solution: The linear programming model follows.

$$\begin{aligned} \text{Maximize:} \quad & Z = 100x_1 + 150x_2 \\ \text{subject to:} \quad & 8,000x_1 + 4,000x_2 \leq 40,000 \\ & 15x_1 + 30x_2 \leq 200 \\ & x_1, x_2 \geq 0 \quad \text{and integer} \end{aligned}$$

where

x_1 = number of presses

x_2 = number of lathes

The decision variables in this model are restricted to whole machines. The fact that both decision variables, x_1 and x_2 , can assume any integer value greater than or equal to zero is what gives this model its designation as a total integer model.

Optimal Solution: $x_1 = 1, x_2 = 6, Z = 1000$

This solution indicates that if the machine shop owner purchases one press and six lathes, a daily increase in profit of \$1,000 will result.

3. Solve the following linear programming model using the branch and bound method.

$$\begin{array}{ll}\text{Minimize:} & Z = 3x_1 + 6x_2 \\ \text{subject to:} & 7x_1 + 3x_2 \leq 10 \\ & x_1, x_2 \geq 0 \quad \text{and integer}\end{array}$$

4. Consider the following linear programming model

$$\begin{array}{ll}\text{Maximize:} & Z = 5x_1 + 4x_2 \\ \text{subject to:} & 3x_1 + 4x_2 \leq 10 \\ & x_1, x_2 \geq 0 \quad \text{and integer}\end{array}$$

- a. Solve this model using the branch and bound method.
 - b. Demonstrate the solution partitioning graphically.
5. A tailor makes wool tweed sport coats and wool slacks. He is able to get a shipment of 150 square yards of wool cloth from Scotland each month to make coats and slacks, and he has 200 hours of his own labor to make them each month. A coat requires 3 square yards of wool and 10 hours to make, and a pair of pants requires 5 square yards of wool and 4 hours to make. He earns \$50 in profit from each coat he makes and \$40 from each pair of slacks. He wants to know how many coats and slacks to produce to maximize profit.
- a. Formulate an integer linear programming model for this problem.
 - b. Determine the integer solution to this problem using the branch and bound method. Compare this solution with the solution without integer restrictions and indicate if the rounded-down solution would have been optimal.
6. A jeweler and her apprentice make silver pins and necklaces by hand. Each week they have 80 hours of labor and 36 ounces of silver available. It requires 8 hours of labor and 2 ounces of silver to make a pin, and 10 hours of labor and 6 ounces of silver to make a necklace. Each pin also contains a small gem of some kind. The demand for pins is no more than six per week. A pin earns the jeweler \$400 in profit, and a necklace earns \$100. The jeweler wants to know how many of each item to make each week in order to maximize profit.

- a. Formulate an integer programming model for this problem.
 - b. Solve this model using the branch and bound method. Compare this solution with the solution without integer restrictions and indicate if the rounded-down solution would have been optimal.
7. The owner of the Consolidated Machine Shop has \$10,000 available to purchase a lathe, a press, a grinder, or some combination thereof. The following integer linear programming model has been developed for determining which of the three machines (lathe, x_1 ; press, x_2 ; grinder, x_3) should be purchased in order to maximize the annual profit.

$$\begin{aligned}
 \text{Maximize:} \quad & Z = 1000x_1 + 700x_2 + 800x_3 \quad (\text{profit}, \$) \\
 \text{subject to:} \quad & 5000x_1 + 6000x_2 + 4000x_3 \leq 10000 \quad (\text{cost}, \$) \\
 & x_1, x_2, x_3 = 0 \text{ or } 1
 \end{aligned}$$

Solve this model using the branch and bound method.

4.4 Binary or 0-1 variables

- Binary variables are employed when there is a YES or NO situation. That is, to indicate whether a selection is made or not. For example, suppose we have four different projects to consider. We can either *select* a project or *not select* a project. The decision variables can therefore be defined as follows:

$$\begin{aligned}
 x_1 &= 1 && \text{if project 1 is selected; 0 otherwise.} \\
 x_2 &= 1 && \text{if project 2 is selected; 0 otherwise.} \\
 x_3 &= 1 && \text{if project 3 is selected; 0 otherwise.} \\
 x_4 &= 1 && \text{if project 4 is selected; 0 otherwise.}
 \end{aligned}$$

Or simply, $x_i = 1$ if project i is selected; 0 otherwise, where $i = 1, 2, 3, 4$.

- Now suppose each project has a lifespan of three months (January, February and March), with corresponding outlays (or costs) in thousands of dollars as shown in Table ???. Also, suppose the funds available for selected projects each month and net returns in thousands of dollars from each project are as indicated.

Table 1: Projects

Project	1	2	3	4	Funds
January	58	44	26	23	120
February	25	29	13	17	80
March	43	25	23	29	95
Return	217	125	88	109	

- In this case, our objective is to maximize returns; that is,

$$\text{Maximize } 217x_1 + 125x_2 + 88x_3 + 109x_4$$

- Since project outlays (or costs) are constrained by available funds, we have the following constraint inequalities:

$$58x_1 + 44x_2 + 26x_3 + 23x_4 \leq 120$$

$$25x_1 + 29x_2 + 13x_3 + 17x_4 \leq 80$$

$$43x_1 + 25x_2 + 23x_3 + 29x_4 \leq 95$$

- Also, the decision variables must be binary i.e., $x_i = 0 - 1$.
- The **final model** is therefore given as:

$$\begin{aligned} & \text{Maximize, } 217x_1 + 125x_2 + 88x_3 + 109x_4 \\ & \text{subject to :} \\ & \quad 58x_1 + 44x_2 + 26x_3 + 23x_4 \leq 120 \\ & \quad 25x_1 + 29x_2 + 13x_3 + 17x_4 \leq 80 \\ & \quad 43x_1 + 25x_2 + 23x_3 + 29x_4 \leq 95 \\ & \quad x_i = 0 - 1(\text{binary}) \end{aligned}$$

- Upon solving the above model using Excel's Solver (see Figure ??), we obtain:

$$x_1 = 1, x_2 = 0, x_3 = 1 \quad \text{and} \quad x_4 = 1.$$

That is, to maximize net returns, the company should undertake projects 1,3 and 4 only, with a corresponding net return of 414 thousand dollars.

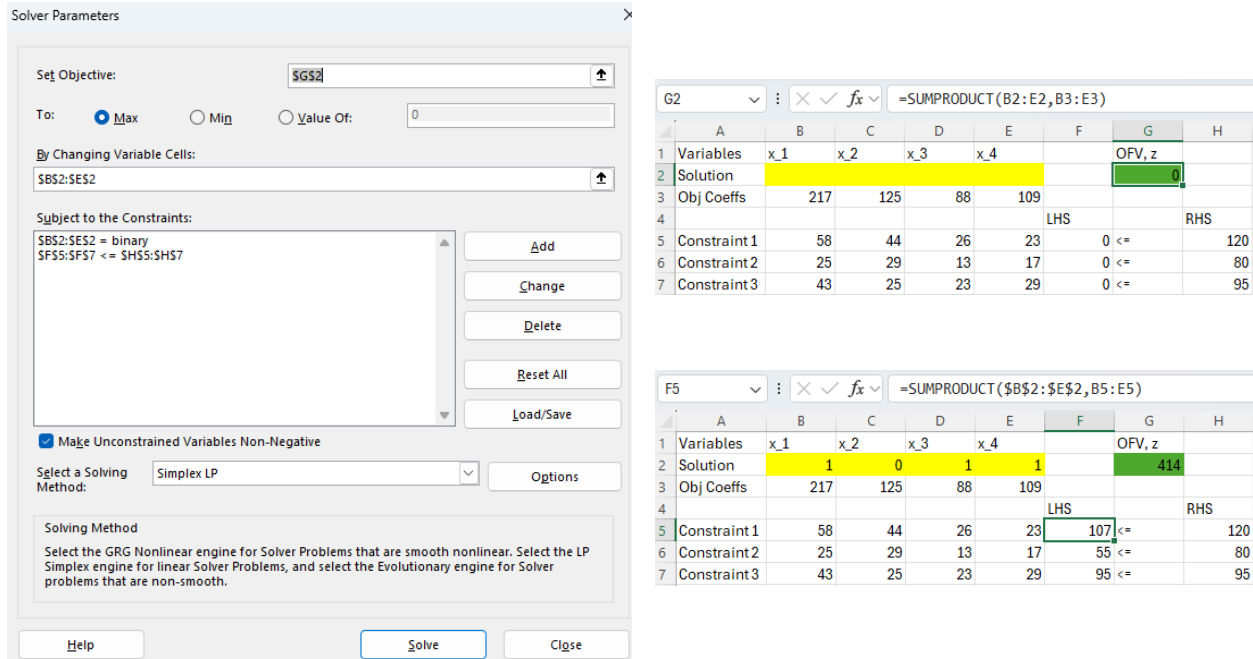


Fig. 17: Solution of Project selection problem

4.4.1 Fixed Cost Problem

- Suppose a small company receives an order to supply 1000 units of a product. The company has three machines that can be used to produce the products. Here are the variable costs per unit produced from each machine, the fixed costs and the machine's capacity.

	machine 1	machine 2	machine 3
Variable cost/unit	2.39	1.99	2.99
Fixed cost	300	250	400
Capacity	400	550	600

- Our objective is to minimize the total costs.
- We will need two types of decision variables in this case. One set for the number of units produced from each machine.

$$x_i = \text{number of units produced on machine, } i, \quad (i=1,2,3,4).$$

That is,

x_1 = Number of units produced on machine 1

x_2 = Number of units produced on machine 2

x_3 = Number of units produced on machine 3

- Since fixed costs indicate that the entire cost will be incurred if the corresponding machine is used to produce it all, we define another set of decision variable.
- For machine 1, we can write $Y_1 = 1$ if machine 1 is used; 0 otherwise.

$$\text{So if } x_1 > 0, Y_1 = 1; \quad \text{otherwise } Y_1 = 0.$$

- For machine 2, we can write $Y_2 = 1$ if machine 2 is used; 0 otherwise.

$$\text{So if } x_2 > 0, Y_2 = 1; \quad \text{otherwise } Y_2 = 0.$$

- For machine 3, we can write $Y_3 = 1$ if machine 3 is used; 0 otherwise.

$$\text{So if } x_3 > 0, Y_3 = 1; \quad \text{otherwise } Y_3 = 0.$$

- For all the three machines, we write

$$Y_i = 1 \quad \text{if machine } i \text{ is used; } 0 \quad \text{otherwise (for } i = 1, 2, 3, 4).$$

- For the objective function, which is to minimize the total costs, we write:

$$\text{Minimize} \quad \underbrace{2.39x_1 + 1.99x_2 + 2.99x_3}_{\text{variable costs}} + \underbrace{300Y_1 + 250Y_2 + 400Y_3}_{\text{fixed costs}}$$

That is, we multiply variable costs by units produced and multiply fixed costs by corresponding 0-1 variables. For example, when $Y_1 = 1$ in $(300Y_1)$, it means that machine 1 is used and the fixed cost of 300 will be incurred; and when $Y_1 = 0$, machine 1 is not used and so the fixed cost of 300 will not be incurred.

- For the constraints, we have an order to supply 1000 units, so we write:

$$x_1 + x_2 + x_3 = 1000.$$

Equality ($=$) is used here because we have to meet the order or demand placed by the customer.

- For the capacities, we normally write

$$x_1 \leq 400; \quad x_2 \leq 550; \quad x_3 \leq 600$$

- The constraint $x_1 \leq 400$ simply means that we can produce upto 400 units on machine 1; but if the machine is not used at all, this will not be possible. So, whenever we have fixed costs associated with minimum requirements or maximum capacity, we usually multiply the capacity by the corresponding binary variables. Thus, the correct format should be: $x_1 \leq 400Y_1$. That is, when $Y_1 = 1$, we can produce upto 400 on machine 1 and when $Y_1 = 0$, 0 units should be produced.

- We do the same to machines 2 and 3. So, $x_2 \leq 550Y_2$ and $x_3 \leq 600Y_3$. Thus,

$$x_1 - 400Y_1 \leq 0,$$

$$x_2 - 550Y_2 \leq 0,$$

$$x_3 - 600Y_3 \leq 0.$$

- The **final model** is therefore given by:

Minimize $2.39x_1 + 1.99x_2 + 2.99x_3 + 300Y_1 + 250Y_2 + 400Y_3$
subject to :

$$x_1 + x_2 + x_3 = 1000,$$

$$x_1 - 400Y_1 \leq 0,$$

$$x_2 - 550Y_2 \leq 0,$$

$$x_3 - 600Y_3 \leq 0,$$

$$x_i \geq 0 \text{ (integer),}$$

$$Y_i = 0 - 1 \text{ (binary).}$$

- Using Excel's Solver (see Figure ??), the solution to the above ILPP is:

$$x_1 = 0, x_2 = 550, x_3 = 450, Y_1 = 0, Y_2 = 1, Y_3 = 1.$$

Showing that only machines 2 and 3 should be used. The minimum total cost is 3090.

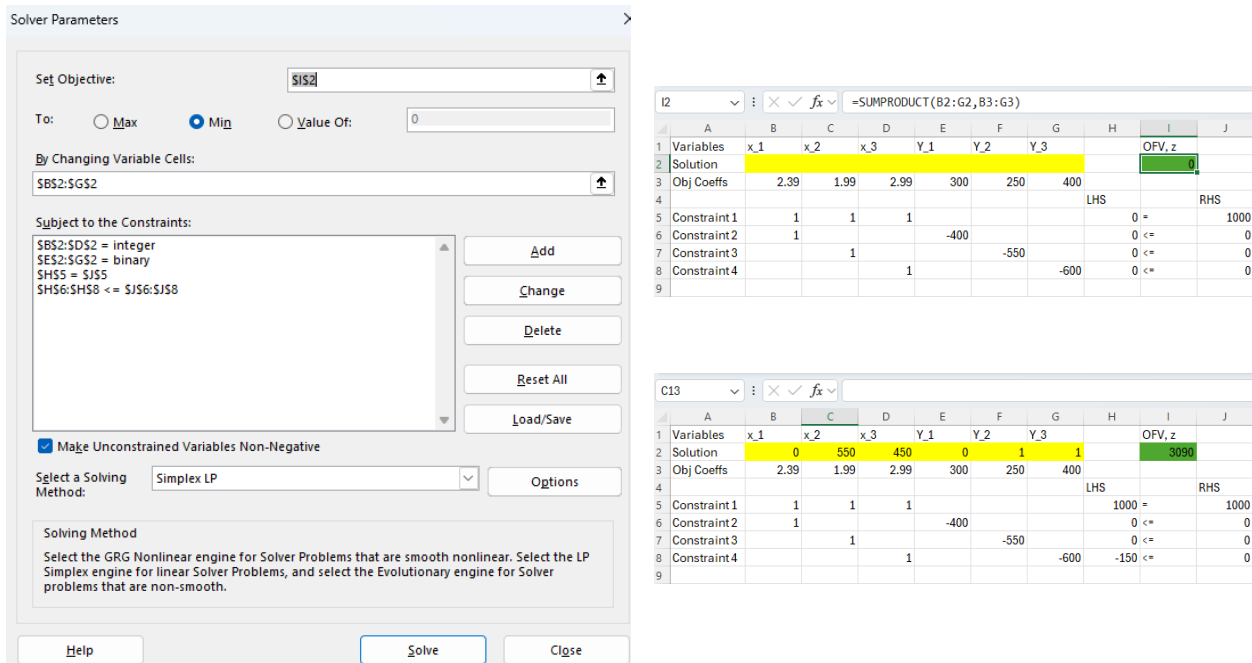


Fig. 18: Solution of Fixed Cost Problem

4.4.2 Logical constraints using Binary or (0-1) integer variables

Suppose a company has 5 projects available to choose from. We can define decision variables as follows:

$$x_i = \begin{cases} 1 & \text{if project } i \text{ is selected} \\ 0 & \text{otherwise} \end{cases} \quad \text{where } i = 1, 2, 3, 4, 5.$$

That is, $x_1 = 1$ if project 1 is selected and 0 if not selected. Now consider the following requirements:

1. If project 3 is selected, project 5 cannot be selected.

In essence, 3 and 5 cannot coexist. So we write

$$x_3 + x_5 \leq 1.$$

That is, we can have only x_3 [1,0] or only x_5 [0,1] or neither [0,0]. This is referred to as a **mutually exclusive constraint**.

2. Exactly one of projects 2 and 3 must be selected.

This is a mutually exclusive case in which we are forced to choose one of projects 2 and 3. So we write

$$x_2 + x_3 = 1.$$

That is, we can have only x_2 [1,0] or only x_3 [0,1]. We cannot have both [1,1] or neither [0,0]. This is often referred to as a **multiple choice constraint**.

3. If project 4 is selected, then project 2 must also be selected.

We write

$$x_4 \leq x_2 \quad \text{or} \quad x_2 \geq x_4$$

This is called **conditional (or contingent) constraint**. This is also written as

$$x_4 - x_2 \leq 0 \quad \text{or} \quad x_2 - x_4 \geq 0.$$

4. If project 1 is selected, project 5 must also be, and vice versa.

In this case, we write

$$x_1 = x_5 \quad \text{or} \quad x_1 - x_5 = 0.$$

This is called a **co-requisite constraint**. That is, we either select both or neither ([1,1] or [0,0]).

5. No more than 3 projects in total may be selected.

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq 3.$$

6. At least 2 of the first 3 projects must be selected.

$$x_1 + x_2 + x_3 \geq 2.$$

7. Project 4 must be selected.

$$x_4 = 1.$$

8. Select project 3 or 5 or both.

$$x_3 + x_5 \geq 1.$$

9. If project 4 is NOT selected, then project 2 must also NOT be selected.

This is another conditional case where 2 depends on 4. So we write:

$$x_2 \leq x_4.$$

10. Project 3 and 4 must be selected together.

This again is a co-requisite case where 3 and 4 must be in or out together. So we write

$$x_3 = x_4.$$

11. Project 2 or 5 must be selected but not both.

In other words, we must select exactly one of 2 and 5. Hence, this is a multiple choice case and so we write:

$$x_2 + x_5 = 1.$$

**** End of Topic Four ****